Seok-Hyung Lee

Curriculum Vitae

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Google Scholar: https://scholar.google.com/citations?user=NURGJAwAAAAJ

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Employment

Sep 2025 – Present

Assistant Professor, Sungkyunkwan University (SKKU)
Department of Quantum Information Engineering, College of Engineering
Suwon-si, Gyeonggi-do, Republic of Korea

Mar 2023 – Sep 2025

Postdoctoral Associate, University of Sydney
Quantum Theory Group, School of Physics
Sydney, New South Wales, Australia
Mentor: Prof. Stephen D. Bartlett

Education

Mar 2017 – Feb 2023

Ph.D. in Physics, Seoul National University
SNU Quantum Information Science Group, Department of Physics and Astronomy
Seoul, Korea
Advisor: Prof. Hyunseok Jeong
Thesis: *Universal Resource-Efficient Topological Measurement-Based Quantum Computing*
Awarded the BK Excellent Thesis Award

Mar 2013 – Feb 2017

Bachelor in Physics, Seoul National University
Seoul, Korea

Mentorship

(2024) Andrew Li, Honours Student at The University of Sydney
Thesis: Decoding twist defects on the color code
(Awarded the University Medal)

Publications

2026

- **[Preprint]** J. M. Drouet, X. C. Kolesnikow, C. K. McLauchlan, G. M. Nixon, **SHL**, D. J. Williamson, S. D. Bartlett, B. J. Brown, R. Harper,
  Error correction on an array of superconducting qubits with defective components, arXiv:2607.12118 (2026).

- [**Preprint**] J. Zen, X. C. Kolesnikow, C. K. McLauchlan, G. M. Nixon, T. R. Scruby, **SHL**, S. D. Bartlett, B. J. Brown, R. Harper,
  Low-valency scalable quantum error correction with a dynamic compass code,
  arXiv:2604.14299 (2026).
- [**Preprint**] **SHL**, X. C. Kolesnikow, J. Zen, E. T. Hockings, C. K. McLauchlan, G. M. Nixon, T. R. Scruby, S. D. Bartlett, R. Harper, B. J. Brown,
  Scalable quantum error correction tailored for a heavy-hex qubit array,
  arXiv:2604.14296 (2026).
- **SHL**, L. H. English, S. D. Bartlett,
  Efficient Post-Selection for General Quantum LDPC Codes, npj Quantum Inf. 12, 96 (2026).
- [**Review**] J. Lee, S. Omkar, Y. S. Teo, **SHL**, H. Kwon, M. S. Kim, H. Jeong,
  Photonic Hybrid Quantum Computing, Newton 2, 100359 (2026).

2025

- [**Preprint**] S. Baek, **SHL**, D. Min, J. Kim,
  SDQC: Distributed Quantum Computing Architecture Utilizing Entangled Ion Qubit Shuttling,
  arXiv:2512.02890 (2025).
- **SHL**, F. Thomsen, N. Fazio, B. J. Brown, S. D. Bartlett,
  Low-Overhead Magic State Distillation with Color Codes, PRX Quantum 6, 030317 (2025).
- **SHL**, A. Li, and S. D. Bartlett,
  Color code decoder with improved scaling for correcting circuit-level noise,
  Quantum 9, 1609 (2025).

2024

- [**Magazine**] **SHL**,
  색 부호를 활용한 결함허용 양자컴퓨팅 (Fault-Tolerant Quantum Computing with the Color Code),
  물리학과 첨단기술 (Physics and High Technology) 33, 17 (2024).
- J. Lee, N. Kang, **SHL**, H. Jeong, L. Jiang, and S.-W. Lee,
  Fault-Tolerant Quantum Computation by Hybrid Qubits with Bosonic Cat Code and Single Photons, PRX Quantum 5, 030322 (2024).

2023

- **SHL** and H. Jeong,
  Graph-theoretical optimization of fusion-based graph state generation,
  Quantum 7, 1212 (2023).
- [**Conference**] H. Jeong, **SHL**, S. Omkar, Y. S. Teo,
  Highly fault-tolerant quantum computing using both discrete and continuous variables of light, in Optica Quantum 2.0 Conference and Exhibition (Optica Publishing Group, 2023) p. QTu4A.6.
- **SHL**, S. Omkar, Y. S. Teo, and H. Jeong,
  Parity-encoding-based quantum computing with Bayesian error tracking,
  npj Quantum Inf. 9, 39 (2023).
- Y. S. Teo, S. Shin, H. Kwon, **SHL**, and H. Jeong,
  Virtual distillation with noise dilution, Phys. Rev. A 107, 022608 (2023).

2022

- S. Omkar, **SHL**, Y. S. Teo, S.-W. Lee, and H. Jeong,
  All-photonic architecture for scalable quantum computing with Greenberger-Horne-Zeilinger States, PRX Quantum 3, 030309 (2022).
- **SHL** and H. Jeong,
  Universal hardware-efficient topological measurement-based quantum computation via color-code-based cluster states, Phys. Rev. Research 4, 013010 (2022).

2021

- **SHL**, S.-W. Lee, and H. Jeong,
  Loss-tolerant concatenated Bell-state measurement with encoded coherent-state qubits for long-range quantum communication, Phys. Rev. Research 3, 043205 (2021).

2020

- S. Choi, **SHL**, and H. Jeong,
  Teleportation of a multiphoton qubit using hybrid entanglement with a loss-tolerant carrier qubit, Phys. Rev. A 102, 012424 (2020).

Software

ldpc-post-selection: Python package for decoding quantum LDPC codes with cluster-based post-selection (2025)

ConcatMatching: Python package implementing the generalised concatenated matching decoder for decoding arbitrary stabiliser codes (2024)

color-code-stim: Python module for simulating and decoding color code circuits (2024)

OptGraphState: Python package for graph-theoretical optimization of fusion-based graph state generation (2023)

Patents

H. Jeong, SHL, Y. S. Teo, S. Omkar
METHOD AND APPARATUS FOR LINEAR OPTICAL QUANTUM COMPUTING,
US Patent 12,596,949 B2 (2026) & KR Patent, App. 1020220120561 (2024)

Research Grants

Ongoing

[Co-PI] Quantum Algorithms for Pluvial Flood Prediction and Sewer Network Optimization in Seoul
(양자 알고리즘 기반 서울시 내수침수 예측 및 예방형 하수관망 최적화),
NRF / 양자컴퓨팅기반양자이득도전연구사업, Grant No. RS-2026-25612364
Role: Co-Principal Investigator
Period: Jul 2026 – Dec 2028
Amount: KRW 2,500M (total project funding)

[PI] Design of Efficient Magic State Factories for Early Fault-Tolerant Quantum Computing
(초기 결함내성 양자컴퓨팅을 위한 효율적인 마법 상태 공장 설계),
NRF / 개인기초연구 (우수연구-신진연구 유형A), Grant No. RS-2026-25476454
Role: Principal Investigator
Period: Mar 2026 – Feb 2029
Amount: KRW 283M

[PI] Development of Performance Improvement Techniques for QLDPC Decoders
(양자 저밀도 패리티 체크 코드 디코더의 성능 개선 기법 개발),
LG Electronics Inc., Grant No. C2026806851
Role: Principal Investigator
Period: Mar 2026 – Dec 2026
Amount: KRW 55M

Presentations

(Date format: YYYY/MM/DD)

Invited talks

- *Soft Output Decoding and Post-Selection for qLDPC Codes*
  2026 YITP QEC Workshop: Fault-Tolerant Logical Processing, Yukawa Institute for Theoretical Physics, Kyoto University, Kyoto, Japan (2026/09/30)
- *Quantum Error Correction: How to Build a Reliable Quantum Computer from Unreliable Qubits*
  Department of Quantum Information Seminar, Yonsei University, Incheon, Korea (2026/09/15)
- *Quantum Error Correction: How to Build a Reliable Quantum Computer from Unreliable Qubits*
  SNU Quantum Science and Technology Forum / IRC Seminar, Seoul National University, Seoul, Korea (2026/09/10)
- *Scalable Quantum Error Correction Tailored for a Heavy-Hex Qubit Array*
  2026 전국대학생양자정보과학컨퍼런스 (QISCA), Kyung Hee University, Global Campus, Yongin, Korea (2026/08/05)
- *Scalable quantum error correction tailored for a heavy-hex qubit array*
  CMQC Alumni Workshop, Center for Macroscopic Quantum Control, Seoul National University, Seoul, Korea (2026/08/04)
- *Advanced Topics in Quantum Error Correction*
  2026 KIAS-KAIST Quantum Information Workshop, Korea Institute of Advanced Study (KIAS), Seoul, Korea (2026/07/28)
- *Soft Output Decoding and Post-Selection for qLDPC Codes*
  QID2026: qLDPC Theory and Application, Korea Institute for Advanced Study (KIAS), Seoul, Korea (2026/07/06)
- *Scalable quantum error correction tailored for a heavy-hex qubit array*
  TartanQEC Workshop 2026, The University of Edinburgh, Edinburgh, UK (2026/06/25)
- *Scalable quantum error correction tailored for a heavy-hex qubit array*
  7th KISTI-SNU Quantum Network Research Center Workshop, KISTI, Daejeon, Korea (2026/05/11)
- *Quantum Error Correction: How to Build a Reliable Quantum Computer from Unreliable Qubits*
  UST Invited Lecture, University of Science and Technology, Korea (2026/05/08, online)
- *양자 오류 정정 입문: 노이즈로부터 양자 정보를 지키는 법*
  SAIT Seminar, Samsung Advanced Institute of Technology, Suwon, Korea (2026/04/27)
- *Efficient Post-selection for General Quantum LDPC codes*
  2026 KPS Spring Meeting, The Korean Physical Society, Daejeon, Korea (2026/04/24)
- *양자 오류 정정 입문: 노이즈로부터 양자 정보를 지키는 법*
  UST Colloquium, Online (2026/03/18)
- *Magic State Cultivation: Toward Efficient Magic State Cultivation*
  2026 Annual Meeting of the Quantum Information Society of Korea, Seoul, Korea (2026/02/26)
- *Fault-Tolerant Quantum Computing with Color Codes*
  KRISS Seminar, Korea Research Institute of Standards and Science, Daejeon, Korea (2026/02/06)
- *Efficient Post-selection for General Quantum LDPC codes*
  QEC Mini Workshop, Yonsei University, Incheon, Korea (2026/01/16)
- *Fault-Tolerant Quantum Computing with Color Codes*
  SAIT Seminar, Samsung Advanced Institute of Technology, Suwon, Korea (2025/12/19)
- *Basics of Quantum Error Correction*
  IBM Quantum Leadership Program Korea, IBM Korea, Seoul, Korea (2025/11/26)
- *양자 오류 정정 입문: 노이즈로부터 양자 정보를 지키는 법*
  Qiskit Fall Fest 2025, Sungkyunkwan University, Suwon, Korea (2025/11/18)

- *색 코드를 활용한 오류 내성 양자 컴퓨팅*
  8th Korea Quantum Information Science Conference (K-QIS), Korea Institute for Advanced Study (KIAS), Seoul, Korea (2025/11/14)
- *광자 기반 오류 내성 양자컴퓨팅의 현황: Fusion-Based Quantum Computing을 중심으로*
  SAIT Seminar, Samsung Advanced Institute of Technology, Suwon, Korea (2025/11/05)
- *Efficient Post-Selection for General Quantum LDPC Codes*
  SNU SQuIS Group Seminar, SNU Quantum Information Science Group, Department of Physics and Astronomy, Seoul National University, Seoul, Korea (2025/11/04)
- *Efficient Post-Selection for General Quantum LDPC Codes*
  31st QISK Regular Workshop, Quantum Information Society of Korea, Seoul, Korea (2025/10/31)
- [**Tutorial Series**] *Toward Fault-Tolerant Photonic Quantum Computing 2*
  SAIT Seminar, Samsung Advanced Institute of Technology, Suwon, Korea, online (Part 1: 2025/06/20, Part 2: 2025/07/18, Part 3: 2025/08/22)
- *Low-Overhead Magic State Distillation with Color Codes*
  IQM QEC Journal Club, IQM Quantum Computers, München, Germany, online (2025/07/10)
- *Challenge of Decoding 2D Color Codes*
  PsiQuantum-USYD Workshop, PsiQuantum, Palo Alto, California, USA (2025/06/11)
- *Low-overhead magic state distillation with color codes*
  KIST Seminar, Center for Quantum Technology, Korea Institute of Science and Technology, Seoul, Korea (2025/05/19)
- *Challenge of Decoding 2D Color Codes*
  Post-QCTiP Workshop, Freie Universität Berlin, Berlin, Germany (2025/04/28)
- *Towards resource-efficient fault-tolerant quantum computing with error correction*
  SKKU Seminar, Qcenter, Sungkyunkwan University, Suwon, Korea (2025/01/02)
- *Low-overhead magic state distillation with color codes*
  KAIST Seminar, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea (2024/12/23)
- *Low-overhead magic state distillation with color codes*
  Joint PsiQuantum-USYD Workshop, University of Sydney, Sydney, Australia (2024/09/25)
- *Graph-theoretical optimization of fusion-based graph state generation*
  Foxconn Quantum Computing Center Weekly Seminar, Hon Hai Quantum Computing Research Center, Taipei, Taiwan, online (2024/05/17)
- [**Tutorial Series**] *Toward Fault-tolerant Photonic Quantum Computing*
  SAIT Seminar, Samsung Advanced Institute of Technology, Suwon, Korea (Part 1: 2024/04/18, Part 2: 2024/06/05 online, Part 3: 2024/07/12 online)
- *Color code decoder with improved scaling for correcting circuit-level noise*
  CMQC Seminar, Center for Macroscopic Quantum Control, Seoul National University, Seoul, Korea (2024/04/15)
- *Color code decoder with improved scaling for correcting circuit-level noise*
  KIST Seminar, Center for Quantum Information, Korea Institute of Science and Technology, Seoul, Korea (2024/04/11)
- *Color code decoder with improved scaling for correcting circuit-level noise*
  KIAS Seminar, School of Computational Sciences, Korea Institute for Advanced Study, Seoul, Korea (2024/04/09)
- *Color code decoder with improved scaling for correcting circuit-level noise*
  Coogee' 24 Sydney Quantum Information Theory Workshop, Sydney, Australia (2024/04/03)
- *Low-overhead Lattice-surgery-based Quantum Computing with the Color Code*
  QST Seminar, Research Institute of Mathematics, Seoul National University, Seoul, Korea (2024/03/15)

- *Pauli-product-measurement-based Quantum Computing with Two-dimensional Color Codes*
  CMQC Seminar, Center for Macroscopic Quantum Control, Seoul National University, Seoul, Korea (2023/09/04)
- *Parity-encoding-based linear-optical quantum computing with graph-theoretical optimization of cluster state generation*
  KIST Workshop on Quantum Information Theory 2022, Center for Quantum Information, Korea Institute of Science and Technology, Seoul, Korea (2022/12/19)
- *Parity-encoding-based linear-optical quantum computing with Bayesian error tracking*
  Quantum Information Science Strategy, KOFST 2022 BrainLink X-Lab Day, Yeosu, Korea (2022/12/15)
- *Universal hardware-efficient topological measurement-based quantum computing via color-code-based cluster states*
  KIST Seminar, Center for Quantum Information, Korea Institute of Science and Technology, Seoul, Korea (2022/05/30)
- *Universal resource-efficient topological measurement-based quantum computation via color-code-based cluster states*
  QST Seminar, Research Institute of Mathematics, Seoul National University, Seoul, Korea (2021/11/12)

Contributed conference talks

- *Scalable Quantum Error Correction Tailored for a Heavy-Hex Qubit Array*
  26th Asian Quantum Information Science Conference (AQIS 2026), KAIST, Daejeon, Korea (2026/08/24)
- [**Rapid Communication**] *Scalable quantum error correction tailored for a heavy-hex qubit array*
  Quantum Error Correction 2026, Santa Barbara, California, USA (2026/06/08)
- *Low-overhead magic state distillation with color codes*
  Quantum Computing Theory in Practice (QCTiP), Berlin, Germany (2025/04/24)
- [**Long talk**] *Color code decoder with improved scaling for correcting circuit-level noise*
  24th Asian Quantum Information Science Conference, Sapporo, Japan (2024/08/26)
- *Linear optical quantum computing tolerant to non-ideal fusions and photon losses*
  15th Asia Pacific Physics Conference, Gyeongju, Korea, online (2022/08/24)
- *Loss-tolerant optical measurement-based quantum computing with incomplete fusion operations*
  Optics and Photonics Congress 2022, Jeju, Korea (2022/07/03)
- *Universal hardware-efficient topological measurement-based quantum computation via color-code-based cluster states*
  33rd Optical Society of Korea Winter Conference, Daejeon, Korea (2022/02/17)

Posters

- *Efficient Post-selection for General Quantum LDPC codes*
  Quantum Error Correction 2026, Santa Barbara, California, USA (2026/06/09)
- *Efficient Post-selection for General Quantum LDPC codes*
  Quantum Information Processing 2026, Riga, Latvia (2026/01/26)
- *Low-Overhead Magic State Distillation with Color Codes*
  7th International Conference on Quantum Error Correction (QEC25), New Haven, Connecticut, USA (2025/08/11)
- *Low-Overhead Magic State Distillation with Color Codes*
  EQUS 2024 Annual Workshop, Noosa Heads, QLD, Australia (2024/12/11)
- *Low-Overhead Lattice-Surgery-Based Quantum Computing with the Color Code*
  Quantum Information Processing 2024, Taipei, Taiwan (2024/01/16)
- *Low-Overhead Lattice-Surgery-Based Quantum Computing with the Color Code*
  EQUS 2023 Annual Workshop, Perth, WA, Australia (2023/11/21)
- *Graph-theoretical optimization of fusion-based graph state generation*
  6th International Conference on Quantum Error Correction (QEC23), Sydney, NSW, Australia (2023/10/31)
- *Graph-theoretical optimization of fusion-based graph state generation*
  Asian Quantum Information Science Conference 2023, Seoul, Korea (2023/08/31)

- *Parity-encoding-based linear optical quantum computing with graph-theoretical optimization*
  Bolder Boulder Quantum Workshop 2023, Boulder, Colorado, USA (2023/06/20)
- *Loss-tolerant all-optical quantum computing architecture using parity-state-encoded multiphoton qubits*
  Quantum Information Processing 2023, Ghent, Belgium (2023/02/06)
- *Loss-tolerant multiphoton-qubit-based linear optical quantum computation with nonideal fusions*
  Single Photon Workshop 2022, Seoul, Korea (2022/11/03)
- *Loss-tolerant linear optical quantum computation with non-ideal fusion operations*
  5th Quantum Information Conference, Seoul, Korea (2022/06/28)
- [**Awarded**] *Universal resource-efficient topological measurement-based quantum computation via color-code-based cluster states*
  21st Asian Quantum Information Science Conference, Tokyo, Japan, online (2021/09/02)
- *Loss-tolerant concatenated Bell-state measurement with coherent-state qubits*
  20th Asian Quantum Information Science Conference, Sydney, Australia, online (2020/12/08)

Awards

(Date format: YYYY/MM/DD)

Graduate Student Research Award
by Department of Physics and Astronomy, Seoul National University (2023/02/24)

Best Student Poster Award
in 21st Asian Quantum Information Science Conference (2021/09/04)

Graduated with Honours
by Department of Physics and Astronomy, Seoul National University (2017/02/24)

Professional Service

- I am a committee member of the **Quantum Optics and Quantum Information Division of the Optical Society of Korea** (2026–present).
- I am an active reviewer of the following journals: *Quantum*, *PRX Quantum*, *Physical Review Letters*, *Physical Review A*, *Quantum Information Processing*, and *IEEE Transactions on Network Science and Engineering*.
- I was a member of the program committee for the **XXXVII IUPAP Conference on Computational Physics (CCP2026)** (2026/08/09–14, Seoul, Korea).
- I was a member of the program committee for the **2026 Annual Meeting of the Quantum Information Society of Korea** (2026/02/25–27, Seoul, Korea).
- I was a member of the scientific organising committee for the **Coogee' 25 Sydney Quantum Information Theory Workshop** (2025/02/10–13, Coogee, Sydney, NSW, Australia).